Naveen Raj B

Principal Engineer | Solution-Oriented | Scalable Systems & LLMs

Puducherry, IN

bnaveenraj1903@gmail.com

Skills

Core Competencies

Distributed Systems, Real-Time Data Platforms, Cloud-Native Architecture, LLM Integration, Observability, Infrastructure Automation

Programming & Frameworks

Python, JavaScript, FastAPI, React, LangChain, OpenAI API, Kafka, WebSockets, Celery, NATS.io, Tailwind, Shadcn, Tantivy

Infrastructure & DevOps

Kubernetes, Docker, Apache Airflow, Ansible, KEDA (Kubernetes Event-Driven Autoscaling), Dask Clusters, Nginx, Traefik, Azure, Docker Compose

Data & Storage

PostgreSQL, MongoDB, Qdrant (Vector DB), Dask, Polars, Parquet, RabbitMQ, Redis

Monitoring & Reliability

Prometheus, Grafana, Alertmanager, NodeExporter, Blackbox Exporter, Memray, Tracemalloc

AI/LLM & Integration

Custom Agent Framework, OpenAI, DALL-E, LangChain, OneDrive, Dropbox, SharePoint, Outlook, Slack, RAG, Vector Search, Document Indexing

Languages

English

Tamil

Telugu

Education

PSG College Of Technology
B.E. Mechanical Engineering
8.43
June 2020

Principal Engineer with 4 years of experience architecting and delivering scalable systems, real-time data platforms, and AI-driven applications. Known for a **solution-first** mindset—**technology-agnostic** approach focused on selecting optimal tools for specific problem domains. Proven expertise in distributed computing, orchestration, observability, and cloud-native systems with a strong track record of delivering measurable business impact through pragmatic engineering leadership.

Experience

Avery by GoodGist Inc — **July 2025 - Present**
Principal Engineer — Bengaluru, IN
https://goodgist.com/avery

Platform Architecture

- **End-to-End Automation**: Architected complete product development pipeline from requirement generation to deployment, requiring zero technical expertise from users

- **Multi-Agent System**: Designed specialized AI agents including dedicated coding agents and debugging agents with intelligent handoff mechanisms

- **Universal LLM Integration**: Created unified interface enabling any Large Language Model to function as powerful, specialized agents

- **Intelligent Model Selection**: Implemented sophisticated model routing system where each workflow step dynamically selects optimal models based on task complexity and performance requirements.

Technical Excellence

- **Advanced Context Management**: Developed efficient file caching and context optimization systems for large-scale code generation

- **Feedback Loop Engineering**: Built intuitive, self-correcting feedback mechanisms ensuring task completion with real-time error resolution

- **Auto-Deployment**: Integrated seamless deployment services with automatic project provisioning

- **Real-Time Monitoring**: Implemented file monitoring systems with automated error detection and response

- **Infrastructure Management**: Designed multi-level proxy architecture using Nginx and Traefik for handling both platform and sandbox traffic

- **Network Intelligence**: Created vector-based network error interception system using Traefik JSON logs for proactive issue resolution.

Quality & Performance

- **Code Quality**: Integrated intelligent linting and type checking systems with real-time code analysis

- **Knowledge Retrieval**: Implemented Tantivy-based scoped knowledge retrieval for contextual code generation

- **Containerization**: Engineered Docker-based sandboxes with sophisticated networking handled by Docker and Traefik

- **Event Architecture**: Built NATS-based event service providing seamless user experience across the entire platform

- **Resource Management**: Developed intelligent auto-pause mechanisms preventing excessive agent execution while maintaining objective completion

- **Optimization**: Achieved maximum token efficiency and enhanced quality through heavily optimized context engineering and advanced prompt engineering techniques

- **Custom Framework**: Built proprietary in-house agent framework from ground up, optimized for the platform's specific requirements

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GoodGist Inc — **July 2023 - June 2025**
Principal Engineer — Bengaluru, IN
https://goodgist.com/

Technical Implementation

- **Task Execution Engine**: Architected Celery-based distributed task processing system with RabbitMQ for reliable, scalable task execution

- **Auto-scaling Infrastructure**: Integrated Kubernetes Event-Driven Autoscaling (KEDA) with Celery workers for dynamic resource optimization

- **Content Generation**: Implemented AI-driven content creation pipelines using DALL-E with automated document indexing capabilities

- **Enterprise Integrations**: Developed comprehensive connector framework and toolkits for enterprise systems including OneDrive, Dropbox, SharePoint, Outlook, and Slack

- **Event-Driven Architecture**: Built sophisticated workflow orchestration with real-time status tracking across distributed agentic systems

- **Document Processing**: Engineered scalable document indexing worker groups for Retrieval-Augmented Generation (RAG) at enterprise scale

Experience

Advanced System Features

- **Email-to-Workflow Engine**: Implemented intelligent email parsing and workflow triggering system supporting both scheduled and on-demand execution
- **Intelligent Routing**: Created catch-all inbox AI agent with intent classification for dynamic workflow selection and execution
- **Distributed Processing**: Built Redis-based distributed email processing layer with advanced locking mechanisms and rate-limited queue management
- **Performance Optimization**: Achieved significant memory usage reduction through systematic bottleneck analysis using memray and tracemalloc profiling
- **Dynamic Analytics**: Developed structured data capture system with automated dashboard generation based on LLM-generated JSON data and intelligent post-processing pipelines.

Researchfin.ai by GoodGist Inc — **February 2022 - July 2023**
Software Engineer — Bengaluru, IN

Core Platform Architecture

- **ETL Pipeline Engineering**: Architected and optimized high-throughput ETL pipelines for efficient processing of large-scale financial datasets
- **Real-Time Data Systems**: Designed robust real-time data ingestion infrastructure using WebSockets and Apache Kafka for low-latency stock market data streaming
- **API Architecture**: Built scalable RESTful API services with emphasis on performance, reliability, and developer experience
- **Data Processing Optimization**: Implemented large-scale computational frameworks using batching strategies and Polars for high-performance data analytics

Infrastructure & Operations

- **Orchestration**: Deployed and managed complex workflow orchestration using Apache Airflow and Ansible for automated task scheduling and configuration management
- **Container Orchestration**: Configured production-ready Dask clusters on Kubernetes with intelligent autoscaling and EFS integration for distributed computing workloads
- **Observability**: Engineered comprehensive monitoring and alerting infrastructure using Prometheus, NodeExporter, and Blackbox Exporter with custom metrics and SLAs
- **Automation**: Developed intelligent service recovery mechanisms with Alertmanager, Ansible, and Apache Airflow for zero-downtime operations
- **Data Optimization**: Created optimized Parquet readers with Polars framework, achieving significant performance improvements in data access patterns

LLM Innovation

- **Early AI Adoption**: Implemented structured output generation and conversational models using LangChain and OpenAI APIs before structured outputs became mainstream
- **Developer Experience**: Created intuitive API abstractions for seamless external service integration

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Bajaj Auto Ltd — **Oct 2020 – Jul 2021**
Area Service Manager — Visakhapatnam, IN

- **Operational Excellence**: Enhanced operational efficiency across 37 service outlets, achieving significant throughput improvements through process optimization
- **Quality Transformation**: Elevated 11 outlets to national top-3 rankings for service quality through systematic performance enhancement initiatives
- **Customer-Centric Solutions**: Reduced service complaints by implementing practical, results-driven solutions rather than rigid process enforcement
- **Feedback Systems**: Established comprehensive customer feedback loop (NPS) to continuously align service delivery with real-world customer needs

Professional Mindset & Learning Approach

- **Hands-On Learner:** I excel by exploring, experimenting, and building, learning deeply through real-world problem solving rather than memorizing every concept upfront.
- **Rapid Adapter:** Quickly internalizes new technologies, architectures, and workflows to deliver practical, high-quality solutions.
- **Impact-First Problem Solver:** Tackles complex challenges with a solution-oriented mindset, optimizing systems for performance, reliability, and scalability.
- **Curious & Pragmatic:** Approaches unknown domains with structured learning and experimentation, turning uncertainty into actionable results.